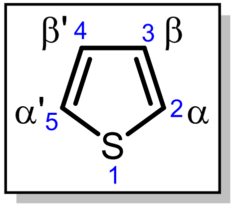
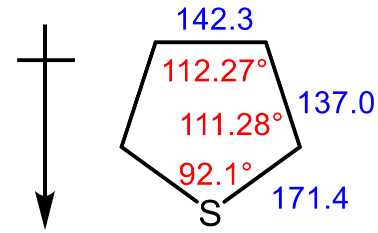
4.3. Thiophen
4.3.1. Struktur



^{1}H-NMR (CS_2)
H-2/5 δ 7.18
H-3/4 δ 6.99

^{13}C-NMR (d^6-Aceton)
C-2/5 δ 125.6
C-3/4 δ 127.3

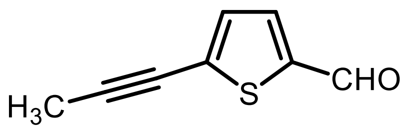


Bindungslängen [pm]
Bindungswinkel

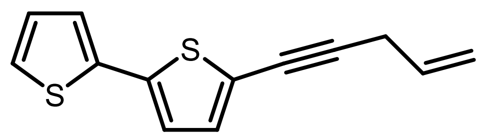
μ = 0.52 D

farblose, geruchslose, wasserunlösliche Flüssigkeit (Schmp.: –38 °C, Sdp.: 84 °C)
Steinkohlenteer, dort: Benzolfraktion (Sdp. Benzol: 80 °C)
Empirische Resonanzenergie: ~ 120 kJ mol^{-1}; Dewar-Resonanzenergie: ~ 27.2 kJ mol^{-1}
$\Rightarrow$ Aromatizität geringer als die des Benzols, aber größer als die des Furans.

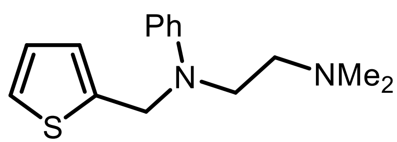
4.3.2. Thiophenhaltige Natur- und Wirkstoffe



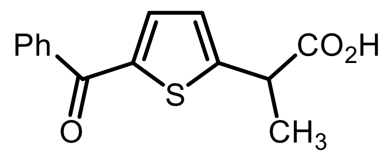
Junipal
aus *Daedelia juniperina*



aus *Echinops sphareocephalus*



Methaphenilen
(Antihistaminikum)

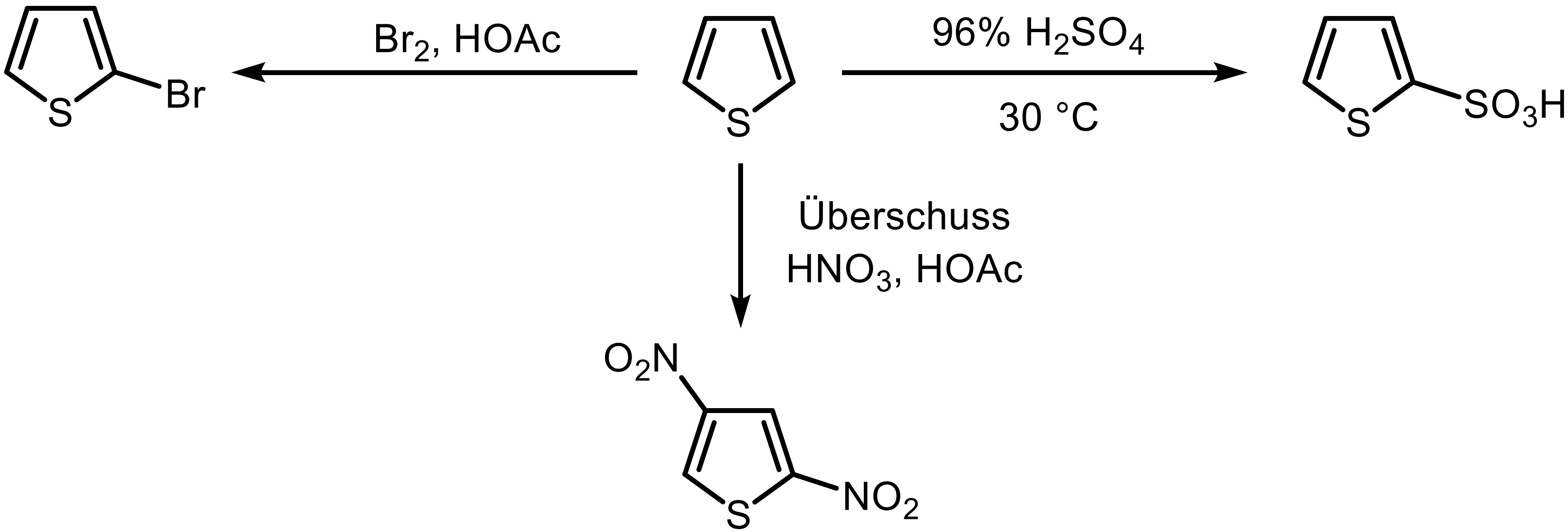


Tiaprofensäure
(Antiphlogistikum)

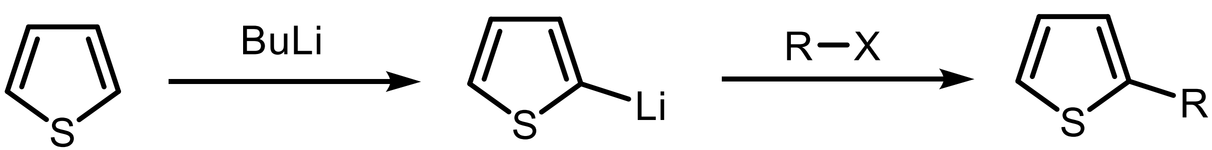
4.3.3. Thiophen-Synthesen
4.3.3.1. Ringtransformationen

Elektrophile Substitutionsreaktionen

44



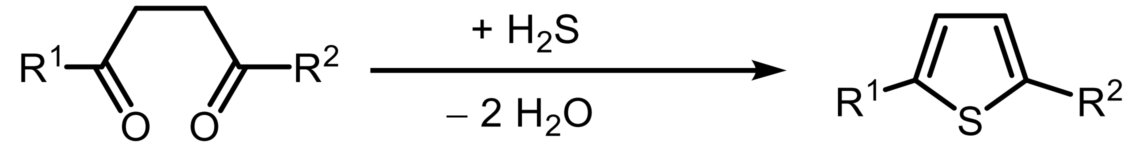
Metallierung



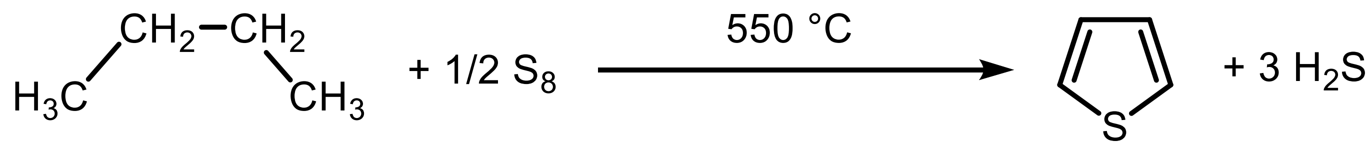
4.3.3.2. Ringaufbauende Synthesen

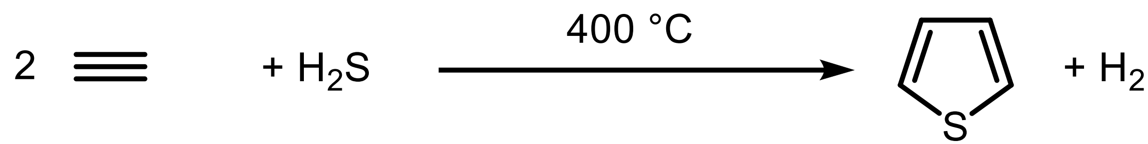
Paal-Synthese

45



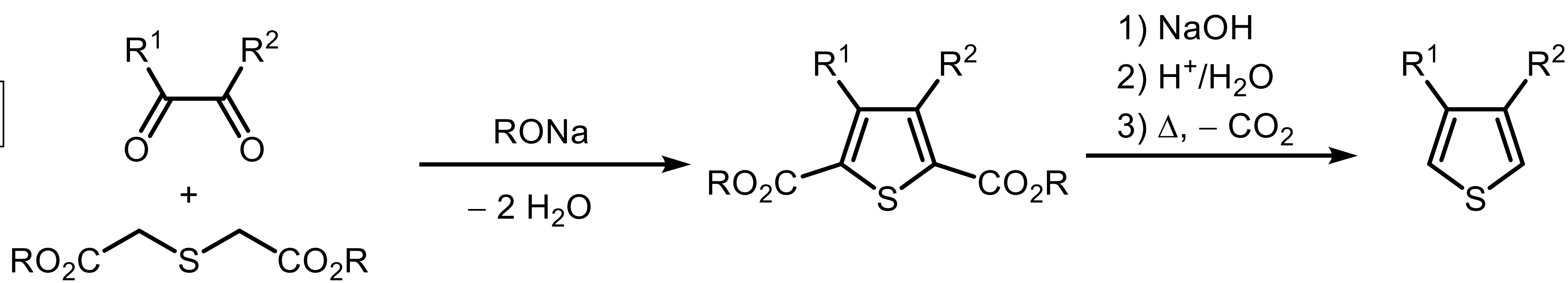
Gasphasen- und Hochdruck-Reaktionen





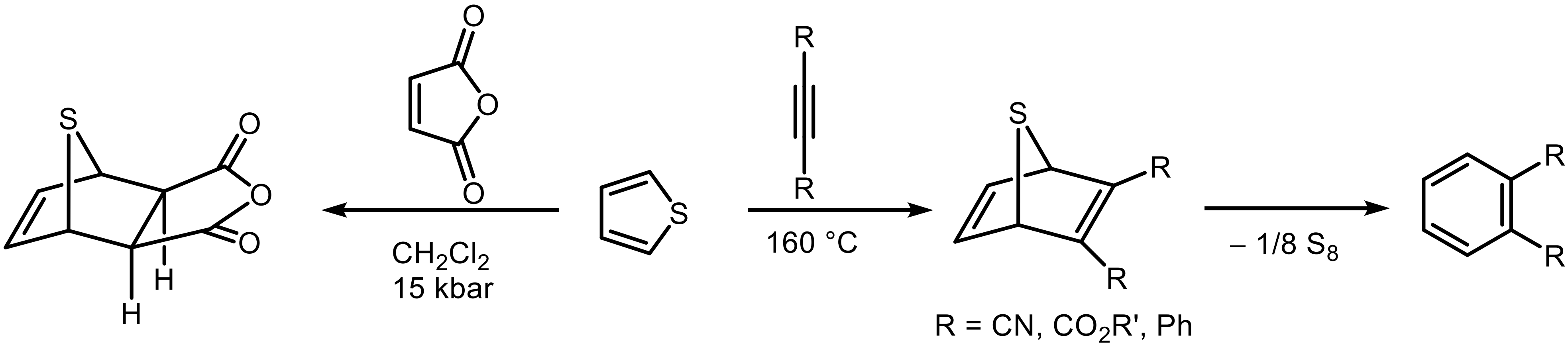
Hinsberg-Synthese

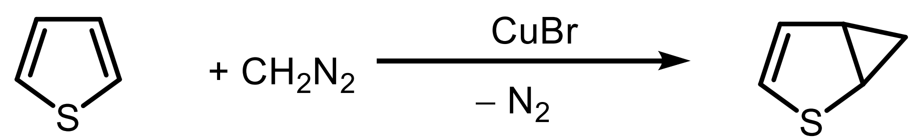
46

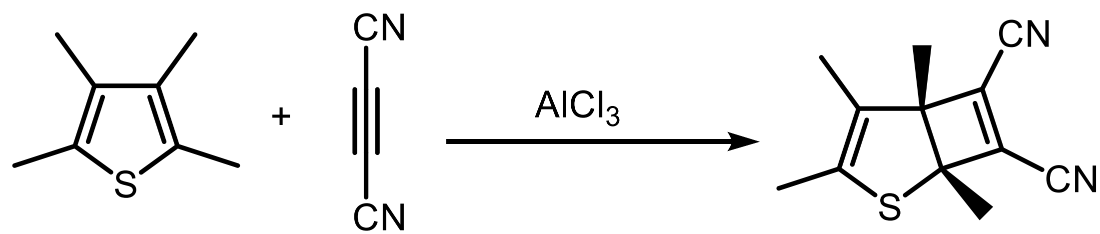


4.3.4. Reaktionen

Additionsreaktionen

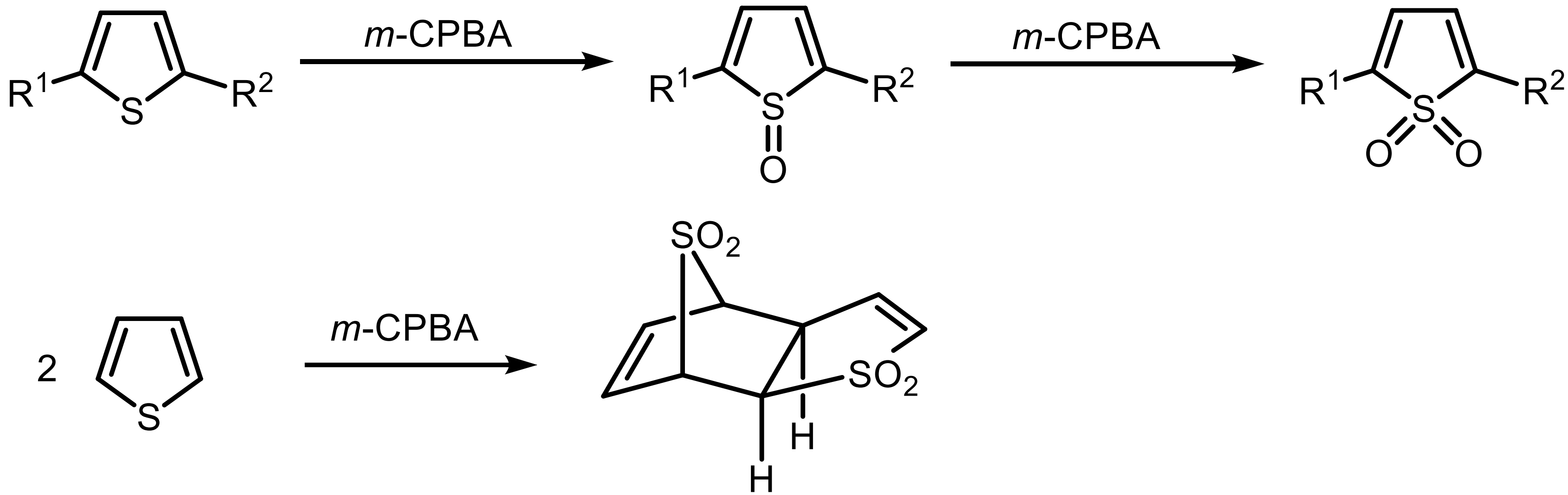




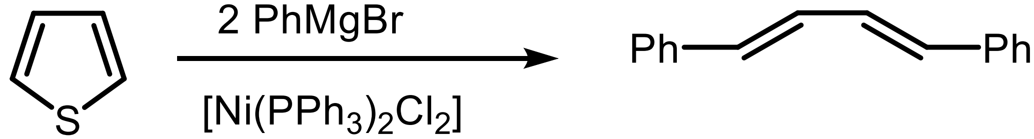


Oxidation

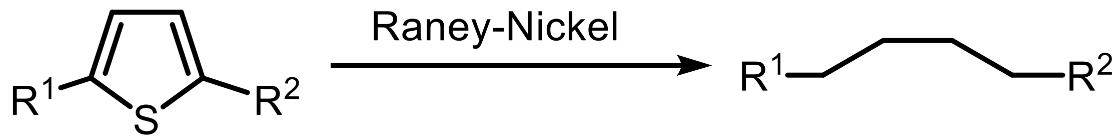
47

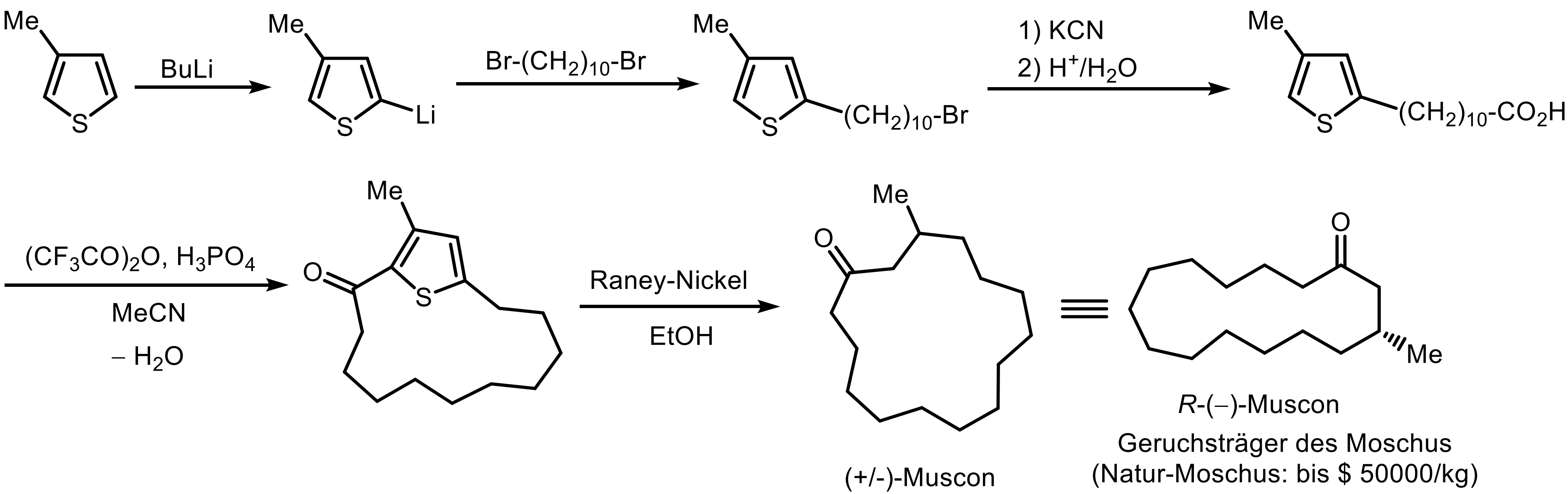


Ringöffnungsreaktionen (reduktive Desulfurierung)



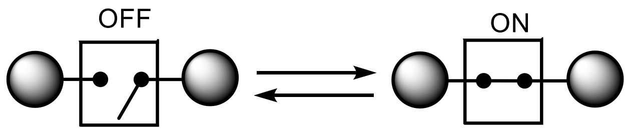
48



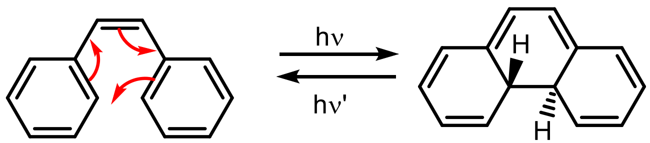


4.3.5. Thiophene in photoschaltbaren Molekülen

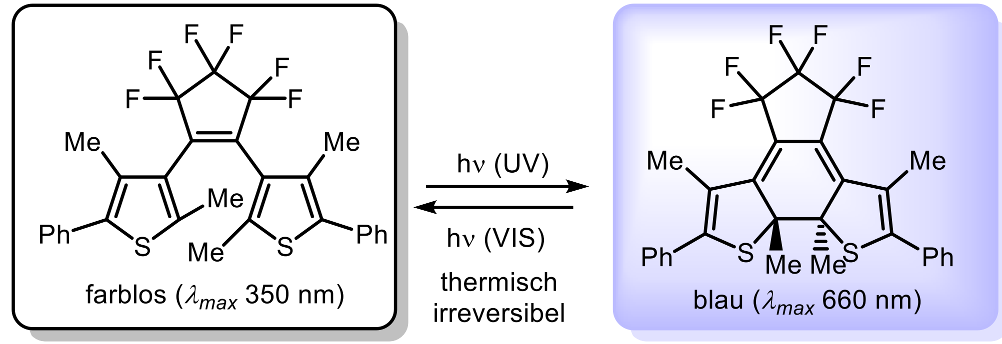
Schalter



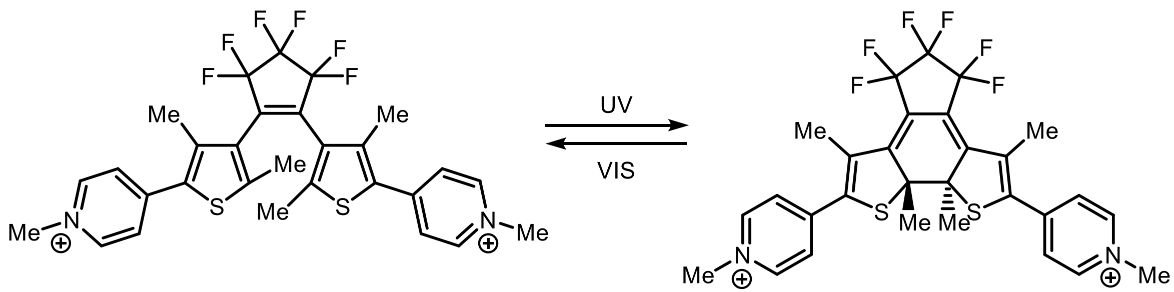
molekularer Schalter



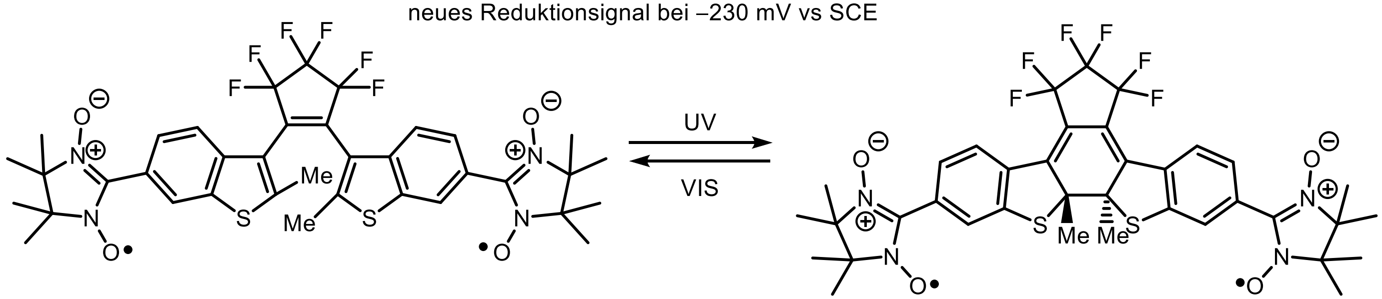
elektrocyclischer con-rotatorischer Ringschluß des *cis*-Stilbens
Nachteil: Dehydrophenanthren ist sehr reaktiv



M. Irie, *Chem. Rev.* **2000**, *100*, 1685



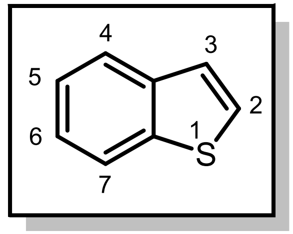
neues Reduktionsignal bei –230 mV vs SCE

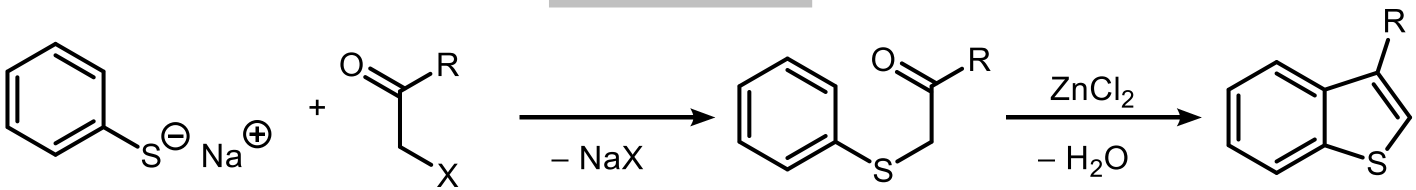


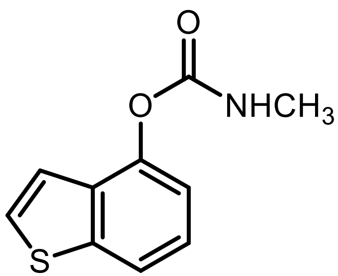
neues antiferromagnetisches Verhalten des Nitronylnitroxid

4.3.6. Benzothiophene

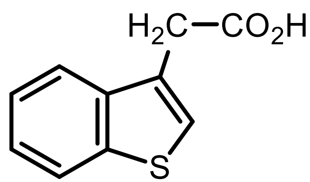
Benzo[*b*]thiophen



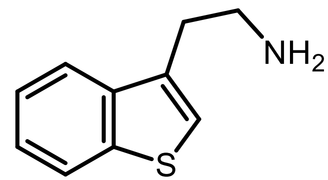




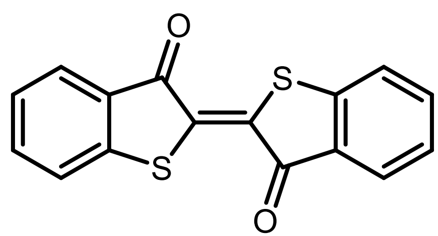
Mobam
(Insektizid)



Benzo[*b*]thiophen-3-ylessigsäure
(Phythohormon)

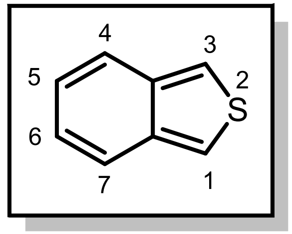


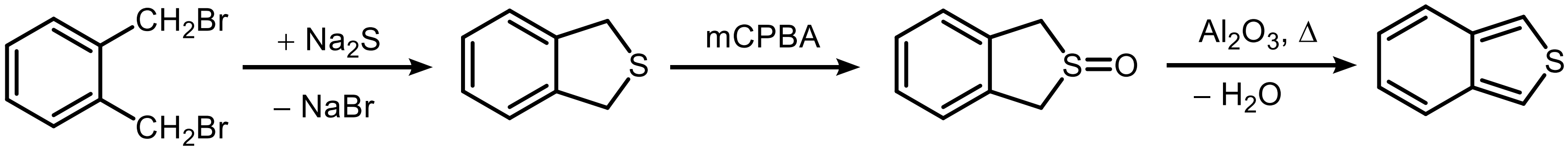
stärkere Wirkung auf das ZNS
als Tryptamin

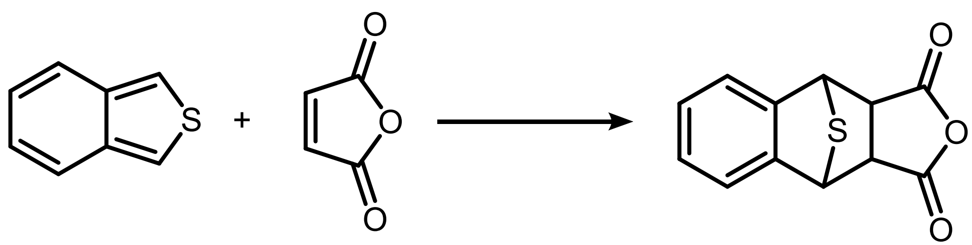


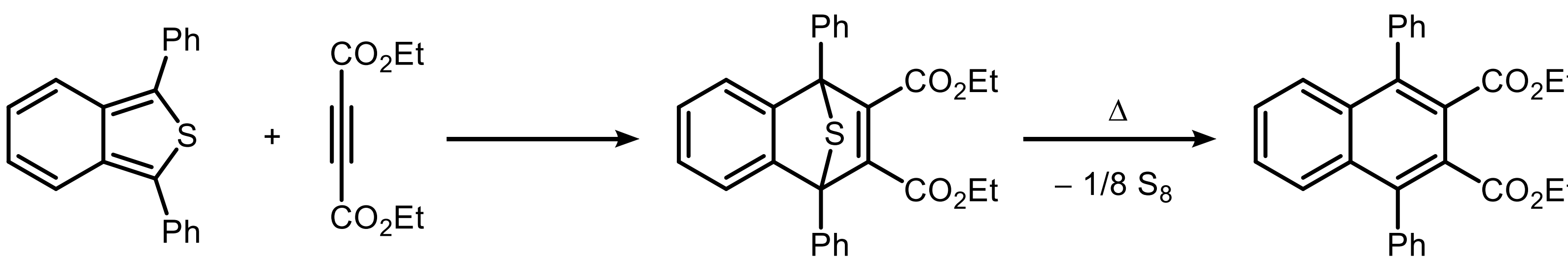
Thioindigo
(rotviolettes Pigment)

Benzo[*c*]thiophen

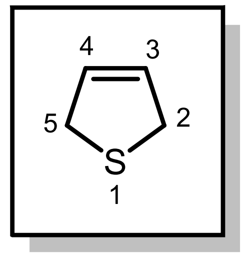


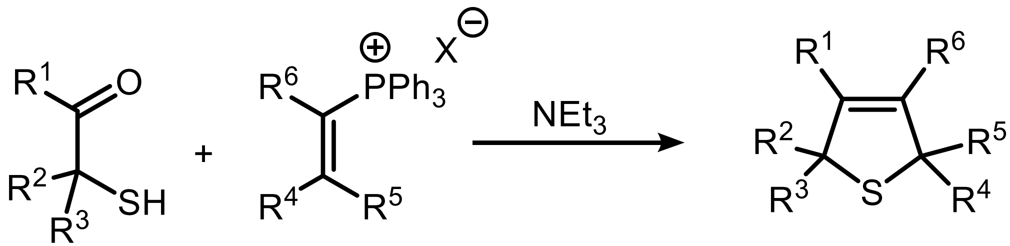




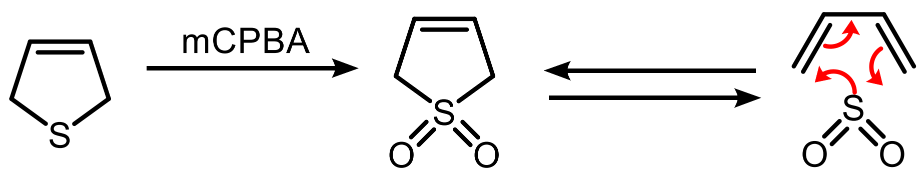


4.3.7. 2,5-Dihydrothiophen

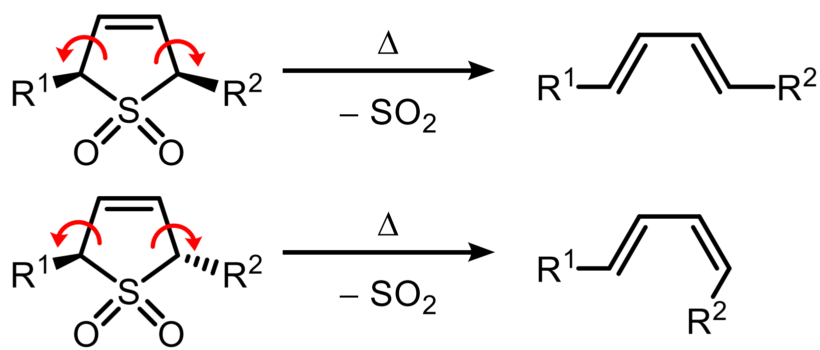




Michael-Addition
und
Wittig-Reaktion



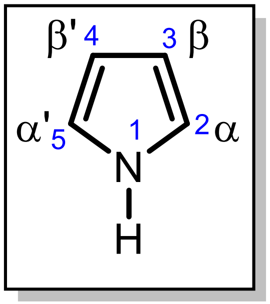
cheletrope Reaktion
(pericyclischer Prozess)

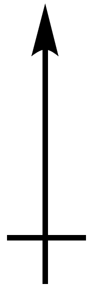


stereospezifisch

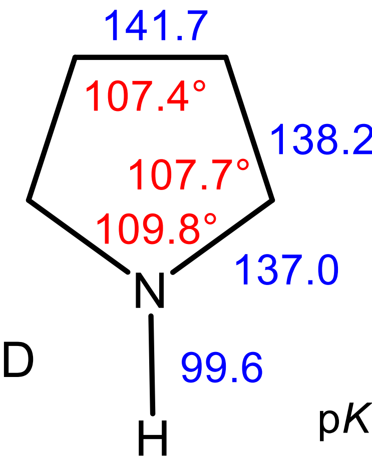
4.4. Pyrrol

4.4.1. Struktur





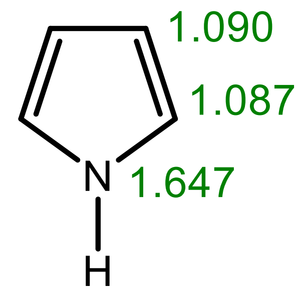
μ = 1.58 D



Bindungslängen [pm]

Bindungswinkel

π-Elektronendichte



pK_S = 17.5 (in Wasser)

Heteroaren	Furan	Pyrrol	Thiophen	Benzol
Heteroatom	O	N	S	-
Elektronegativität	3.5	3.0	2.5	(2.5)
Empirische Resonanzenergie [kJ/mol)	80	100	120	150
Dewar-Resonanzenergie [kJ/mol]	19	22	27	95

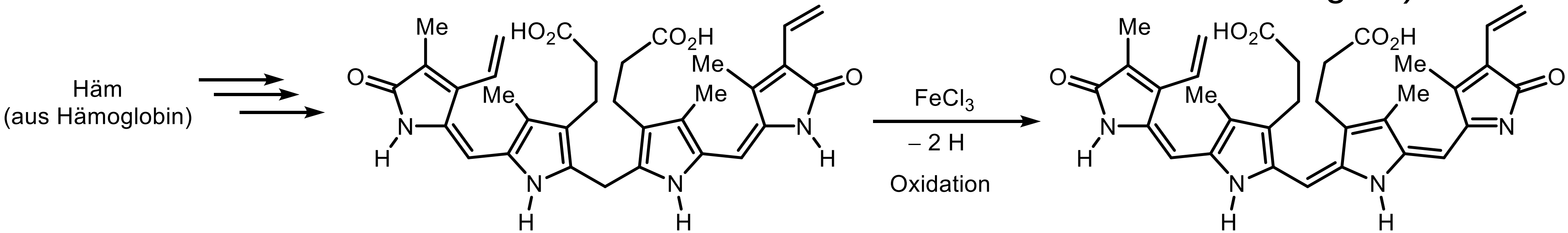
49

Aromatizitätsabstufung: **Furan < Pyrrol < Thiophen** (< Benzol)

4.4.2. Vorkommen

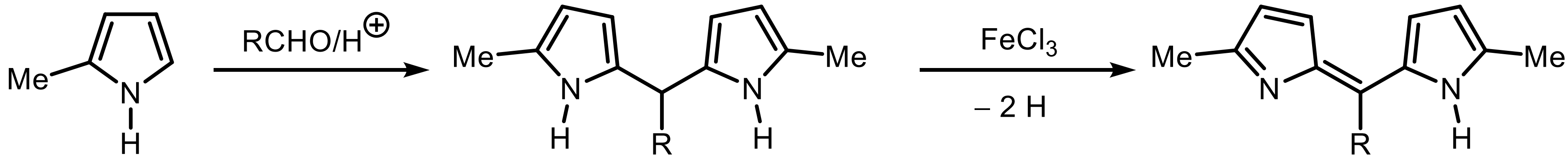
Bilirubin: Abbauprodukt des Blutfarbstoffs in der Galle (lat. "bilis" Galle; "ruber" rot)

Biliverdin (ital./span. "verde" grün)

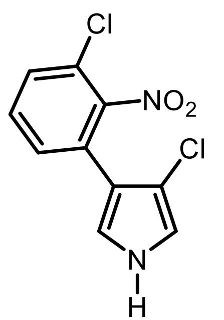


Bilirubin

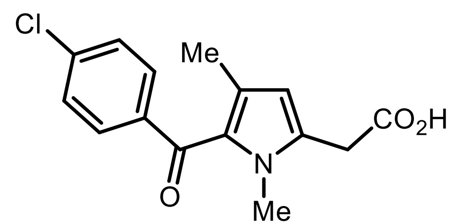
Biliverdin



Pyrrolyl(pyrrol-2-yliden)methan
orange für R = Me



Pyrrolnitrin
(antimykotisch wirksames Antibiotikum)



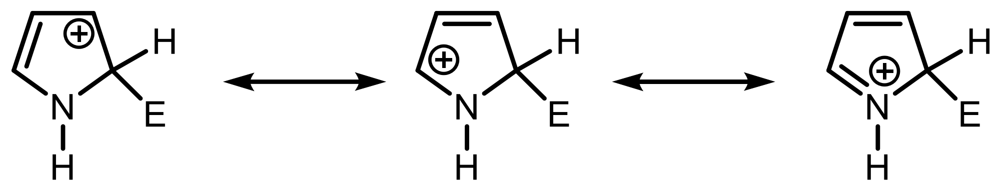
Zomepirac
(COX-Inhibitor, nichtsteroidales entzündungshemmendes Medikament mit fiebersenkender Wirkung, seit 1993 vom Markt wegen allergischer Reaktionen)

4.4.3. Pyrrol-Synthesen
4.4.3.1. Ringtransformationen

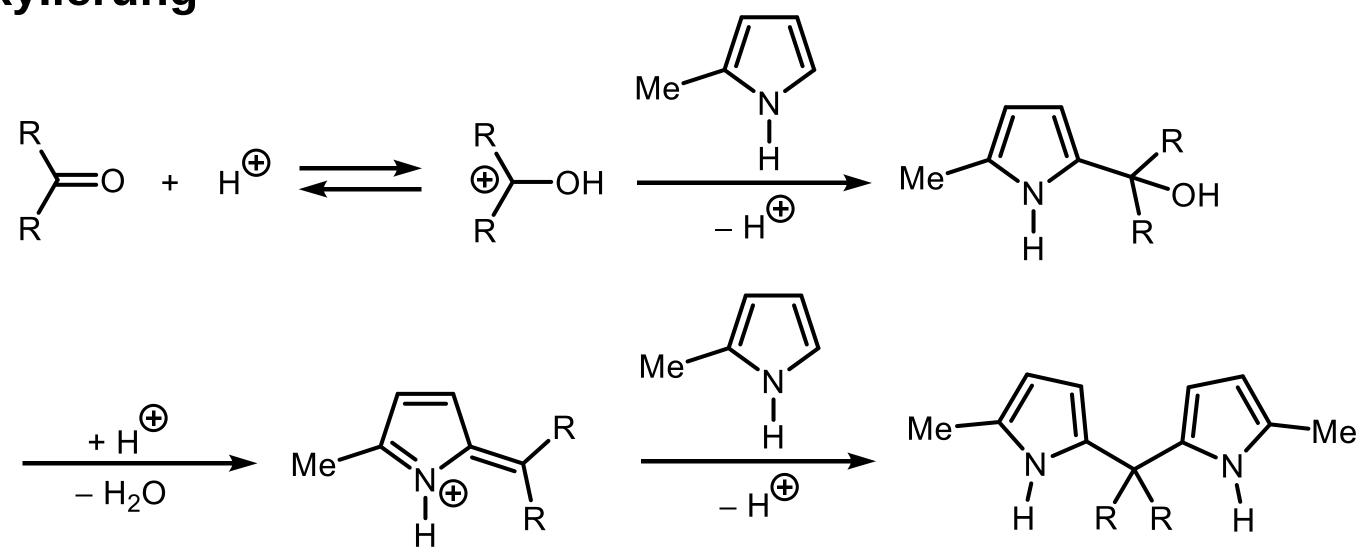
Reaktionen in der α-Position

2-Halopyrrol: durch Reaktion von Pyrrol mit NCS bzw. NBS;
2,3,4,5-Tetrahalopyrrol: durch Reaktion von Pyrrol mit SO_2Cl_2, NaOCl bzw. Br_2;
2-Nitropyrrol: durch Reaktion von Pyrrol mit HNO_3/Ac_2O bei -10 °C;

50

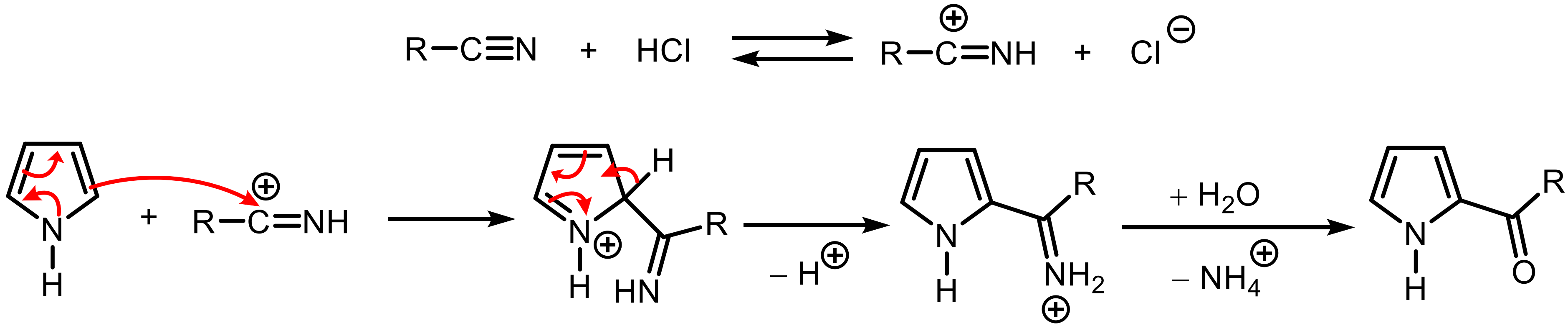


Hydroxyalkylierung

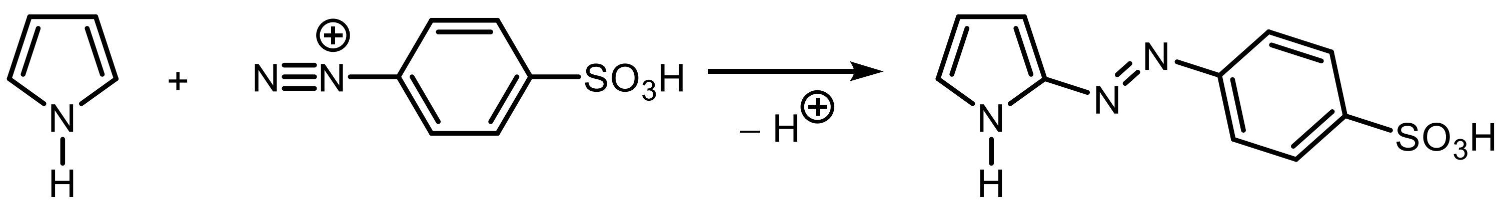


4.4.3.1. Ringtransformationen
Fortsetzung

Houben-Hoesch-Acylierung



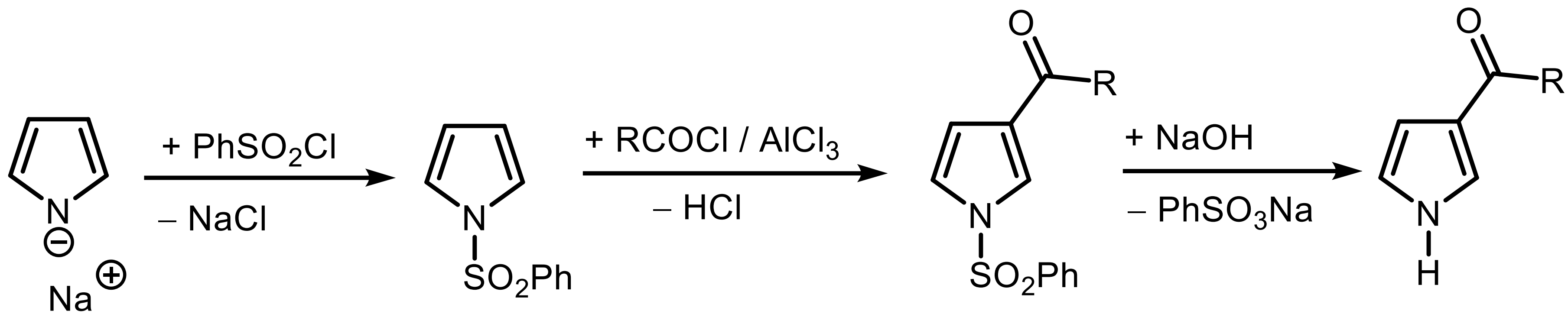
Azokupplung (ebenso: Formylierung, Acylierung)



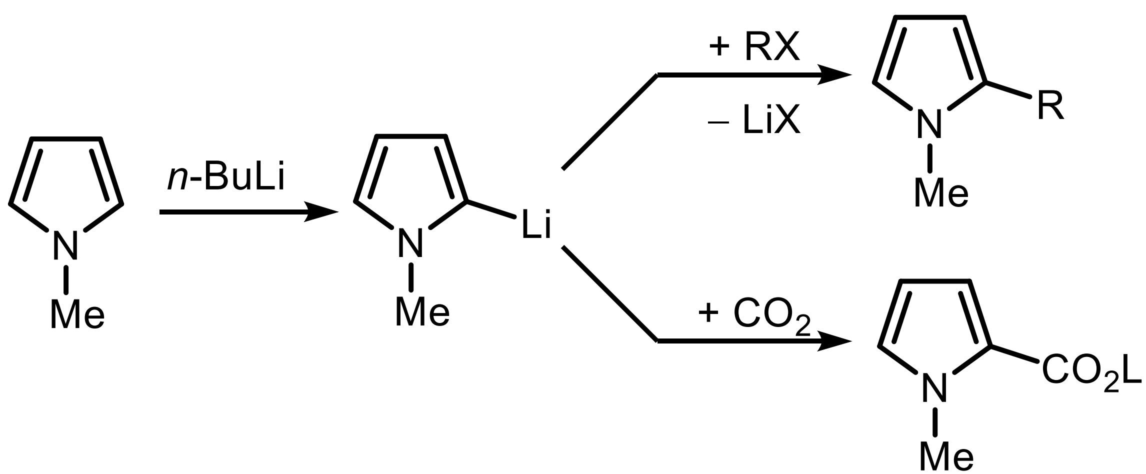
4.4.3.1. Ringtransformationen Fortsetzung

Acylierung der 3-Position: vorher *N*-Sulfonylieren (1. NaH, 2. $PhSO_2Cl$, 3. $RCOCl/(AlCl_3)$, 4. NaOH).

51



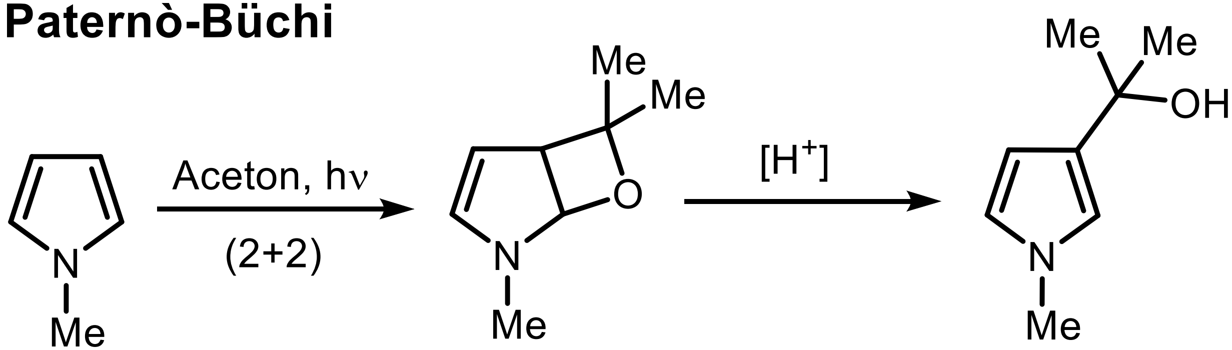
Lithiierung der 2-Position: vorher *N*-Methylieren (1. NaH, 2. MeI, 3. BuLi, 4. z. B. CO_2 zum Li-Salz).



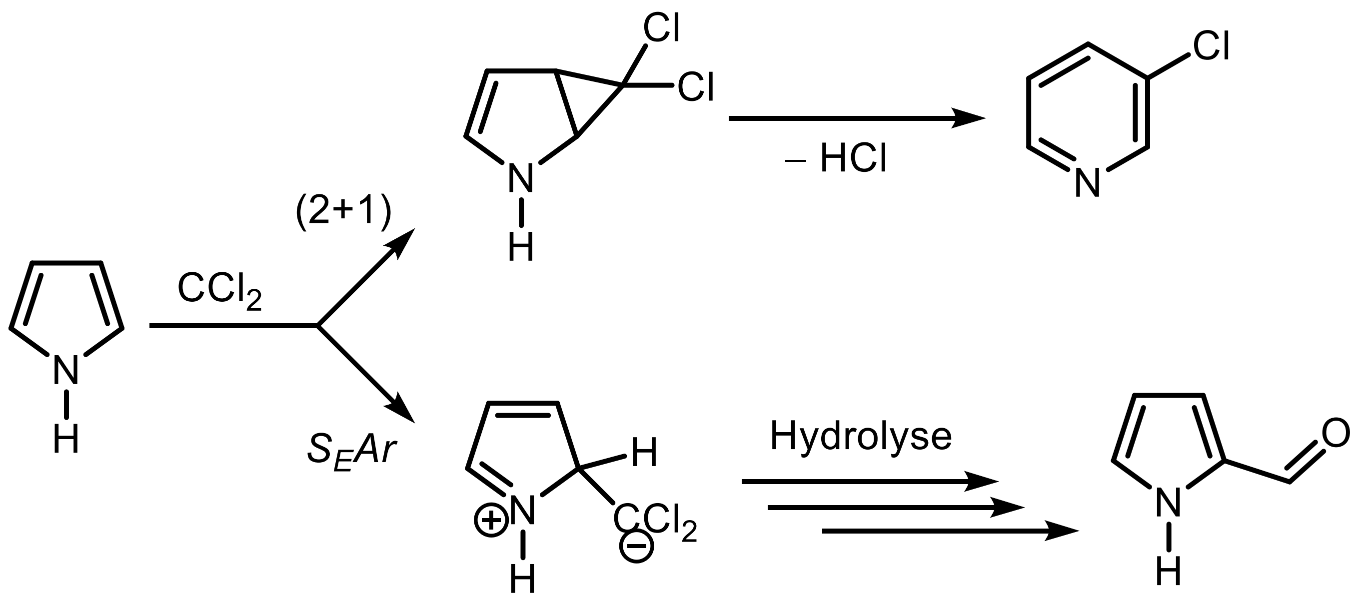
4.4.3.1. Ringtransformationen
Fortsetzung

Cycloadditionen

Paternò-Büchi



Cheletrope (2+1)-Cycloaddition



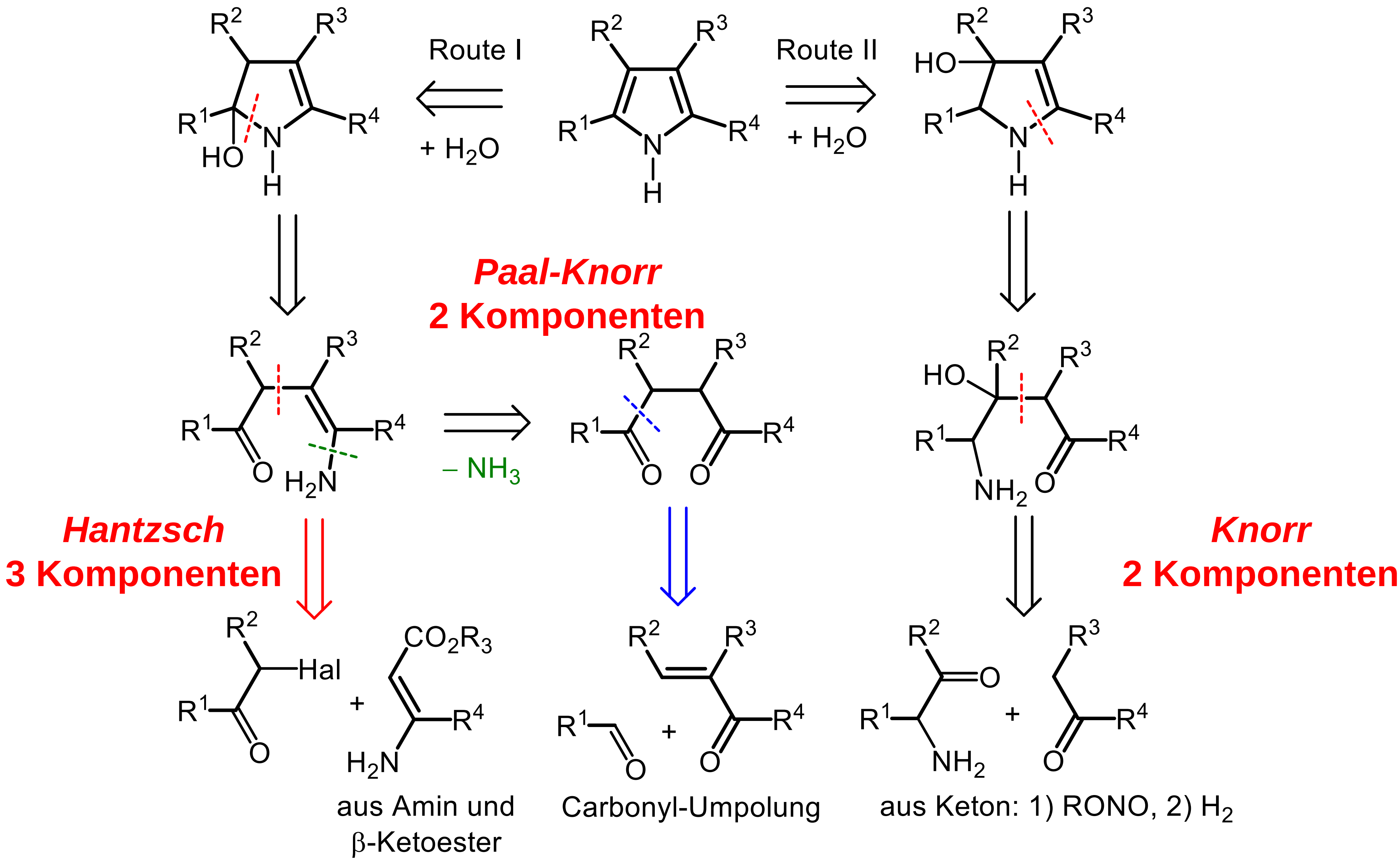
(2+1)-Cycloaddition, wenn CCl_2 aus NaO_2CCCl_3

Reimer-Tiemann, wenn CCl_2 aus $CHCl_3$/NaOH

Reimer-Tiemann-Synthese

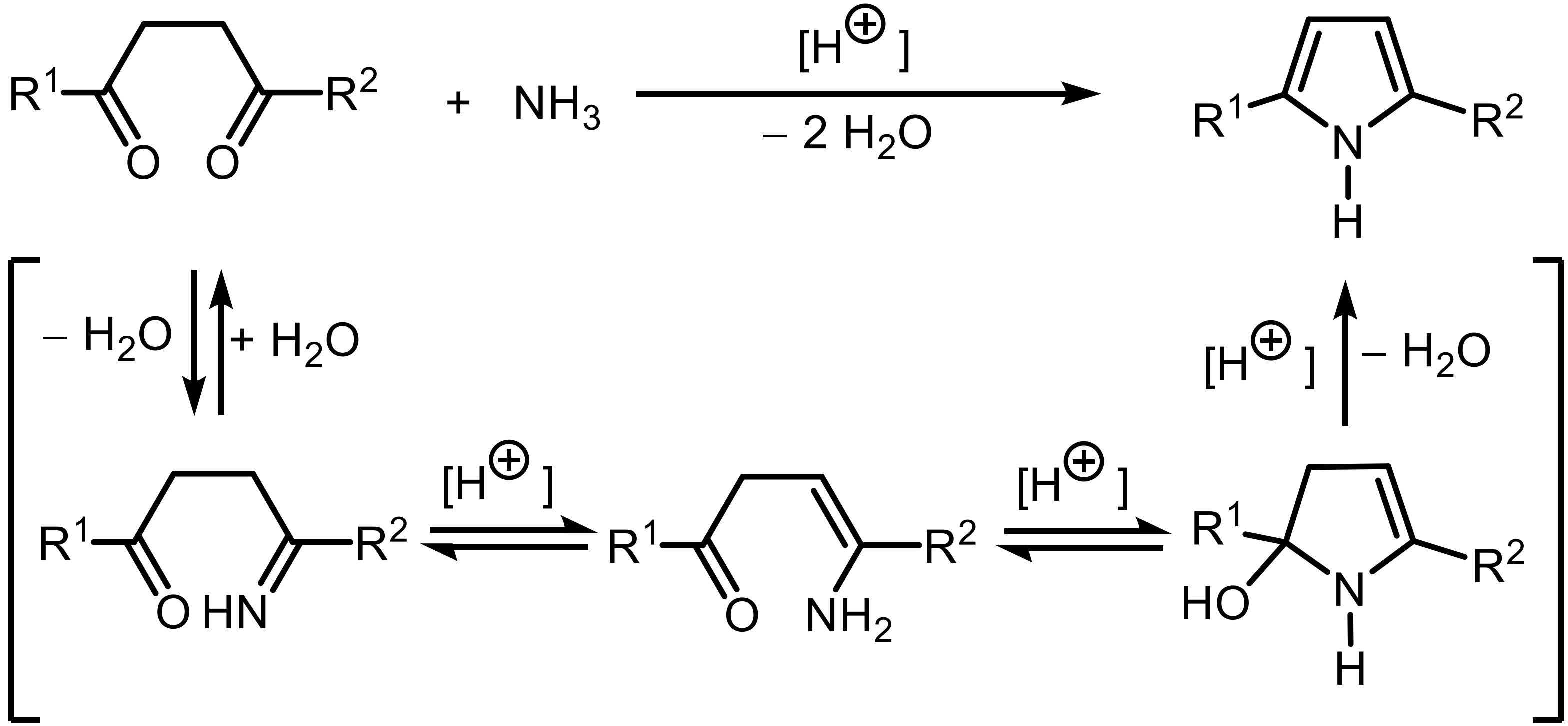
4.4.3.2. Ringaufbauende Synthesen

Retrosynthetische Überlegungen



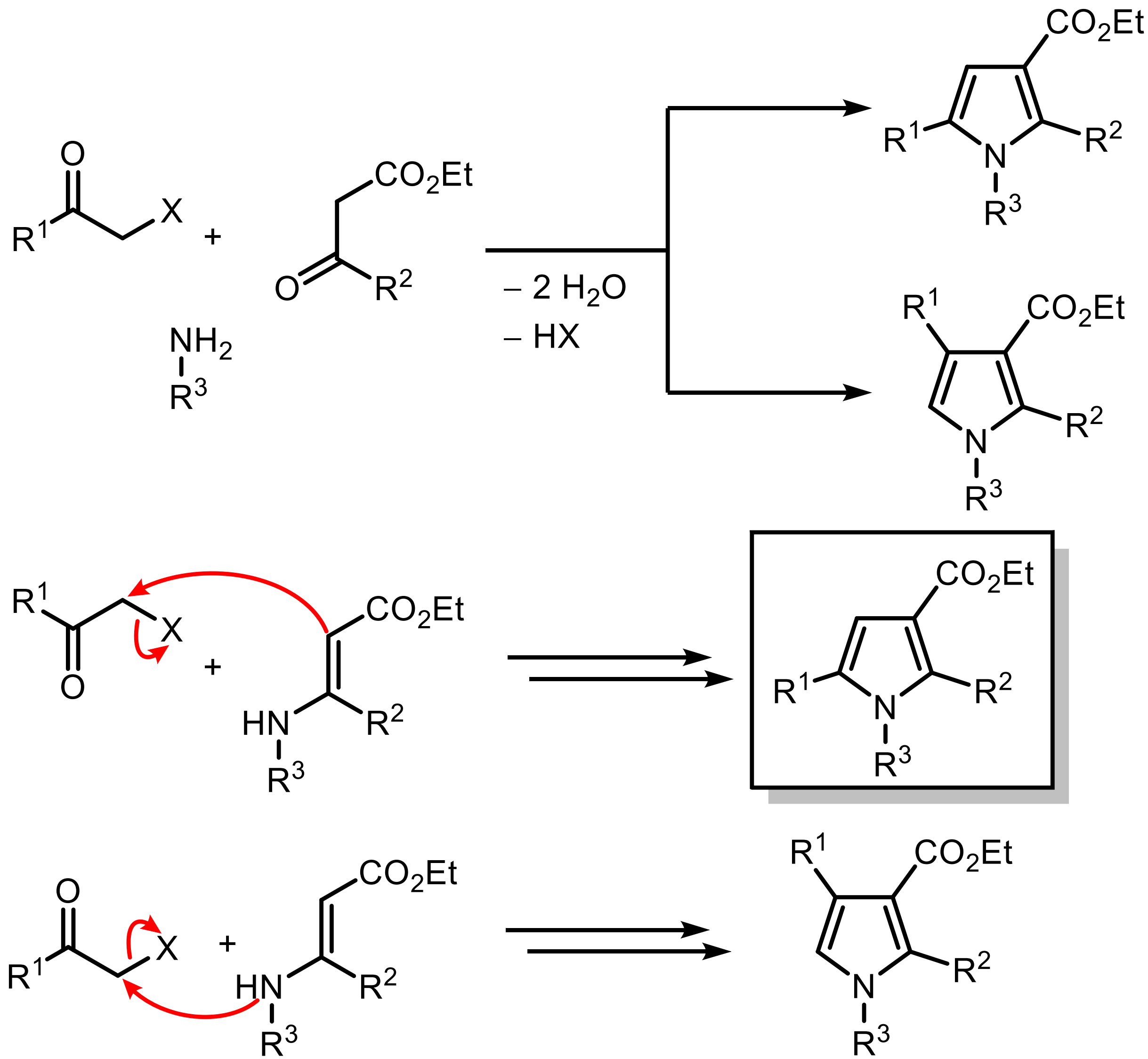
Paal-Knorr-Synthese

52



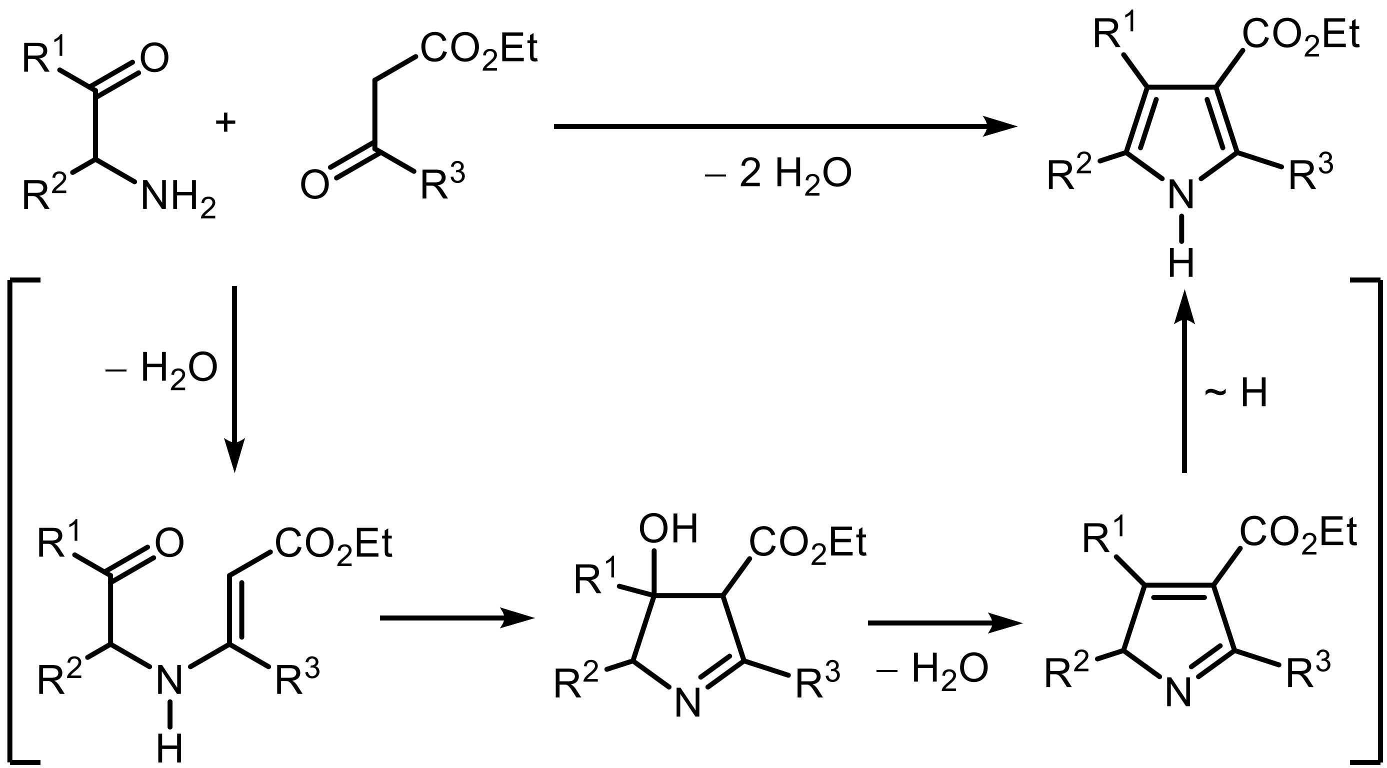
4.4.3.2. Ringaufbauende Synthesen
Hantzsch-Synthese

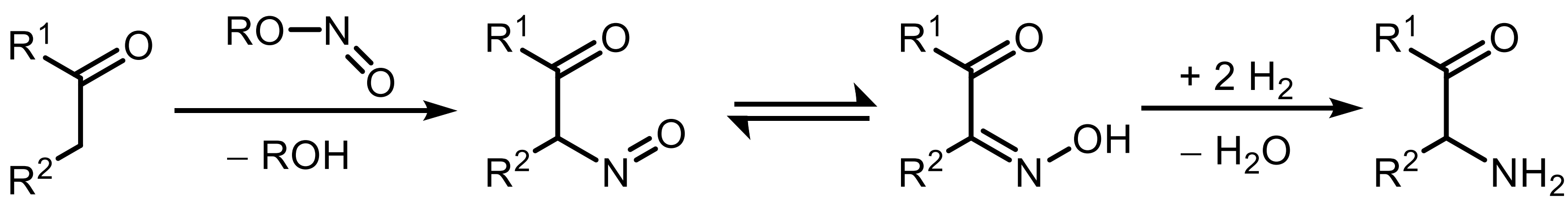
53



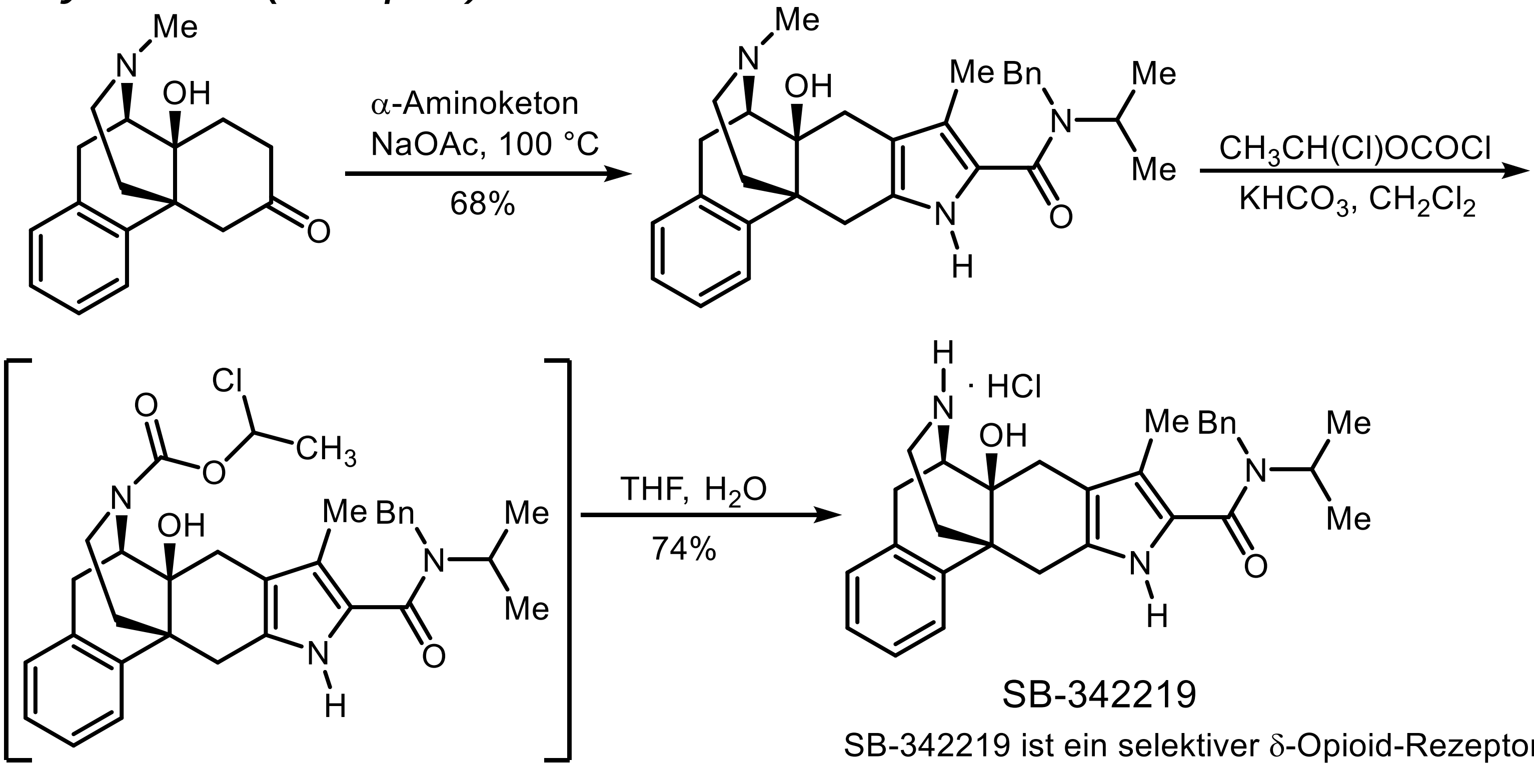
Knorr-Synthese

54





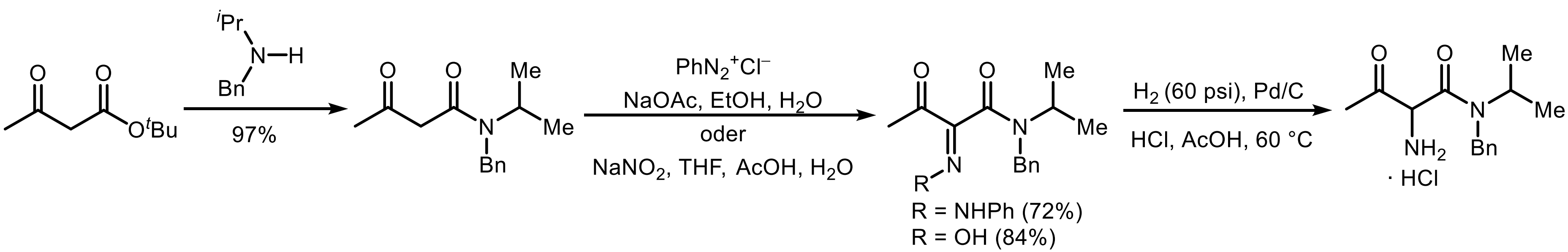
4.4.3.2. Ringaufbauende Synthesen
Knorr-Synthese (Beispiel)



SB-342219

SB-342219 ist ein selektiver δ-Opioid-Rezeptor-Agonist (*Org. Proc. Res. Dev.* **2004**, 279)

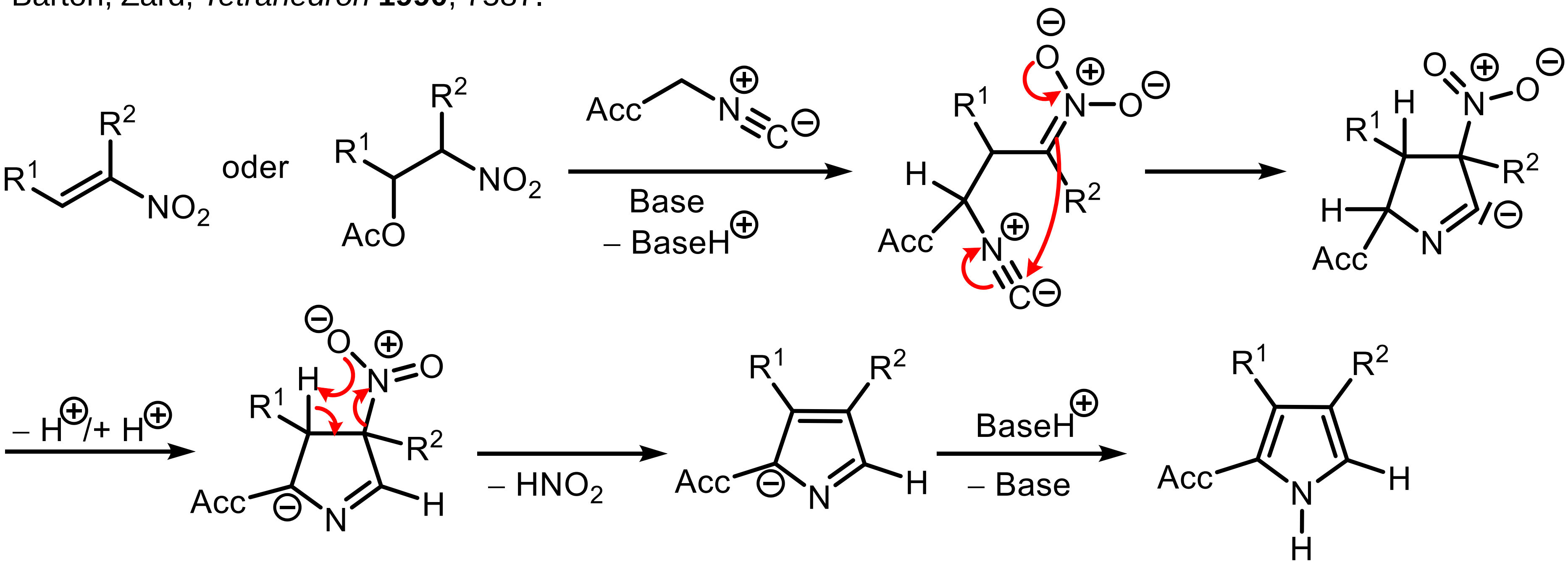
Synthese des α-Aminoketons



4.4.3.2. Ringaufbauende Synthesen

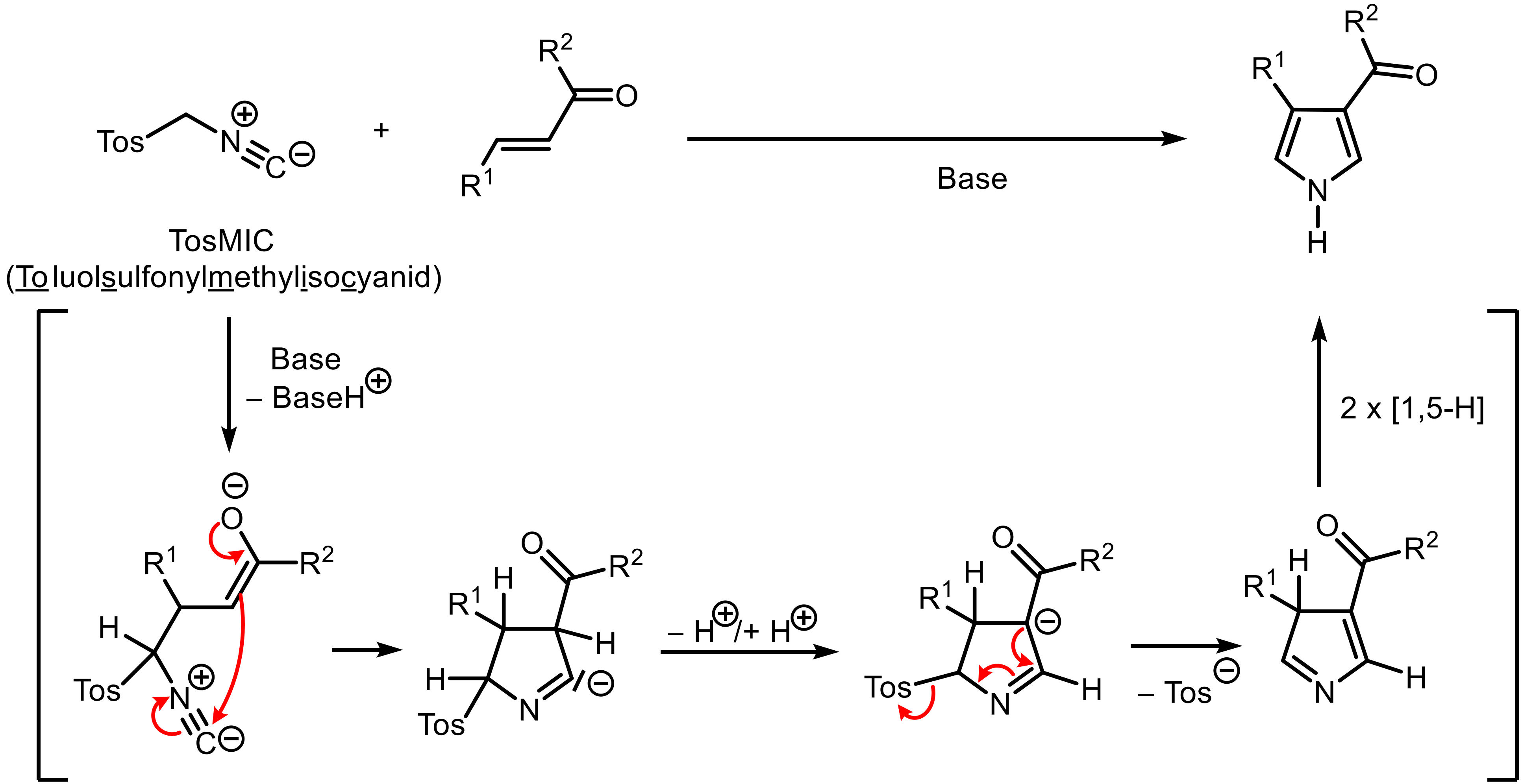
Barton-Zard-Synthese von α-akzeptor-substituierten Pyrrolen

Barton, Zard, *Tetrahedron* **1990**, 7587.



Acc = CO_2R, SO_2Aryl

van-Leusen-Synthese (1972)



4.4.3.2. Ringaufbauende Synthesen

Kenner-Synthese